

Useful Functions

INTERMEDIATE R



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Loads of useful functions

- `sapply()`, `vapply()`, `lapply()`
- `sort()`
- `print()`
- `identical()` ...

Mathematical utilities

```
v1 <- c(1.1, -7.1, 5.4, -2.7)
v2 <- c(-3.6, 4.1, 5.8, -8.0)
mean(c(sum(round(abs(v1))), sum(round(abs(v2)))))
```

abs()

```
v1 <- c(1.1, -7.1, 5.4, -2.7)
v2 <- c(-3.6, 4.1, 5.8, -8.0)
mean(c(sum(round(abs(v1))), sum(round(abs(v2)))))
```

```
abs(c(1.1, -7.1, 5.4, -2.7))
```

```
1.1 7.1 5.4 2.7
```

```
abs(c(-3.6, 4.1, 5.8, -8.0))
```

```
3.6 4.1 5.8 8.0
```

```
mean(c(sum(round(c(1.1, 7.1, 5.4, 2.7))),
        sum(round(c(3.6, 4.1, 5.8, 8.0)))))
```

round()

```
v1 <- c(1.1, -7.1, 5.4, -2.7)
v2 <- c(-3.6, 4.1, 5.8, -8.0)
mean(c(sum(round(abs(v1))), sum(round(abs(v2)))))
```

```
mean(c(sum(round(c(1.1, 7.1, 5.4, 2.7))),
      sum(round(c(3.6, 4.1, 5.8, 8.0)))))
```

```
round(c(1.1, 7.1, 5.4, 2.7))
```

```
1 7 5 3
```

```
round(c(3.6, 4.1, 5.8, 8.0))
```

```
4 4 6 8
```

sum()

```
v1 <- c(1.1, -7.1, 5.4, -2.7)
v2 <- c(-3.6, 4.1, 5.8, -8.0)
mean(c(sum(round(abs(v1))), sum(round(abs(v2)))))
```

```
mean(c(sum(c(1, 7, 5, 3)),
       sum(c(4, 4, 6, 8))))
```

```
sum(c(1, 7, 5, 3))
```

16

```
sum(c(4, 4, 6, 8))
```

22

mean()

```
mean(c(16, 22))
```

19

```
v1 <- c(1.1, -7.1, 5.4, -2.7)
v2 <- c(-3.6, 4.1, 5.8, -8.0)
mean(c(sum(round(abs(v1))), sum(round(abs(v2)))))
```

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Functions for data structures

```
li <- list(log = TRUE,  
          ch = "hello",  
          int_vec = sort(rep(seq(8, 2, by = -2), times = 2)))
```

```
sort(rep(seq(8, 2, by = -2), times = 2)))
```


seq()

```
li <- list(log = TRUE,  
          ch = "hello",  
          int_vec = sort(rep(seq(8, 2, by = -2), times = 2)))
```

```
sort(rep(seq(8, 2, by = -2), times = 2)))
```

```
seq(1, 10, by = 3)
```

```
1 4 7 10
```

```
seq(8, 2, by = -2)
```

```
8 6 4 2
```

rep()

```
li <- list(log = TRUE,  
          ch = "hello",  
          int_vec = sort(rep(seq(8, 2, by = -2), times = 2)))  
sort(rep(c(8, 6, 4, 2), times = 2))
```

```
rep(c(8, 6, 4, 2), times = 2)
```

```
8 6 4 2 8 6 4 2
```

```
rep(c(8, 6, 4, 2), each = 2)
```

```
8 8 6 6 4 4 2 2
```

sort()

```
li <- list(log = TRUE,  
          ch = "hello",  
          int_vec = sort(rep(seq(8, 2, by = -2), times = 2)))
```

```
sort(c(8, 6, 4, 2, 8, 6, 4, 2))
```

```
2 2 4 4 6 6 8 8
```

```
sort(c(8, 6, 4, 2, 8, 6, 4, 2), decreasing = TRUE)
```

```
8 8 6 6 4 4 2 2
```

sort()

```
sort(rep(seq(8, 2, by = -2), times = 2))
```

```
2 2 4 4 6 6 8 8
```

str()

```
li <- list(log = TRUE,  
          ch = "hello",  
          int_vec = sort(rep(seq(8, 2, by = -2), times = 2)))  
  
str(li)
```

```
List of 3  
 $ log   : logi TRUE  
 $ ch    : chr "hello"  
 $ int_vec: num [1:8] 2 2 4 4 6 6 8 8
```

is.*(), as.*()

```
is.list(li)
```

TRUE

```
is.list(c(1, 2, 3))
```

FALSE

```
li2 <- as.list(c(1, 2, 3))  
is.list(li2)
```

TRUE

is.*(), as.*()

```
unlist(li)
```

```
log    ch  int_vec1  int_vec2  ...  int_vec7  int_vec8  
"TRUE" "hello"    "2"    "2"    ...    "8"    "8"
```

append(), rev()

```
str(append(li, rev(li)))  
str(rev(li))
```

```
List of 3  
$ int_vec: num [1:8] 2 2 4 4 6 6 8 8  
$ ch      : chr "hello"  
$ log     : logi TRUE
```


append(), rev()

```
str(append(li, rev(li)))
```

```
List of 3
$ int_vec: num [1:8] 2 2 4 4 6 6 8 8
$ ch      : chr "hello"
$ log     : logi TRUEstr(append(li, rev(li)))
List of 6
$ log     : logi TRUE
$ ch      : chr "hello"
$ int_vec: num [1:8] 2 2 4 4 6 6 8 8
$ int_vec: num [1:8] 2 2 4 4 6 6 8 8
$ ch      : chr "hello"
$ log     : logi TRUE
```

Let's practice!

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Regular Expressions

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Regular Expressions

- Sequence of (meta)characters
- Pattern existence
- Pattern replacement
- Pattern extraction
- `grep()`, `grepL()`
- `sub()`, `gsub()`

grepl()

```
animals <- c("cat", "moose", "impala", "ant", "kiwi")
```

```
grepl(pattern = <regex>, x = <string>)
```

```
grepl(pattern = "a", x = animals)
```

```
TRUE FALSE TRUE TRUE FALSE
```

grepl()

```
grepl(pattern = "^a", x = animals)
```

```
FALSE FALSE FALSE TRUE FALSE
```

```
grepl(pattern = "a$", x = animals)
```

```
FALSE FALSE TRUE FALSE FALSE
```

?regex

grep()

```
animals <- c("cat", "moose", "impala", "ant", "kiwi")
```

```
grepl(pattern = "a", x = animals)
```

```
TRUE FALSE TRUE TRUE FALSE
```

```
grep(pattern = "a", x = animals)
```

```
1 3 4
```

grep()

```
which(grepl(pattern = "a", x = animals))
```

```
1 3 4
```

```
grep(pattern = "^a", x = animals)
```

```
4
```


sub(), gsub()

```
animals <- c("cat", "moose", "impala", "ant", "kiwi")
```

```
sub(pattern = <regex>, replacement = <str>, x = <str>)
```

```
sub(pattern = "a", replacement = "o", x = animals)
```

```
"cot" "moose" "impola" "ont" "kiwi"
```

```
gsub(pattern = "a", replacement = "o", x = animals)
```

```
"cot" "moose" "impolo" "ont" "kiwi"
```

sub(), gsub()

```
animals <- c("cat", "moose", "impala", "ant", "kiwi")
```

```
sub(pattern = "a", replacement = "o", x = animals)
```

```
"cot"  "moose"  "impola" "ont"  "kiwi"
```

```
gsub(pattern = "a", replacement = "o", x = animals)
```

```
"cot"  "moose"  "impolo" "ont"  "kiwi"
```

sub(), gsub()

```
gsub(pattern = "a|i", replacement = "_", x = animals)
```

```
"c_t"  "_mp_l_" "_nt"  "k_w_"
```

```
gsub(pattern = "a|i|o", replacement = "_", x = animals)
```

```
"c_t"  "m__se"  "_mp_l_" "_nt"  "k_w_"
```

Let's practice!

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Times & Dates

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Today, right now!

```
today <- Sys.Date()  
today
```

```
"2015-05-07"
```

```
class(today)
```

```
"Date"
```

```
now <- Sys.time()  
now
```

```
"2015-05-07 10:34:52 CEST"
```

```
class(now)
```

```
"POSIXct" "POSIXt"
```

Create Date objects

```
my_date <- as.Date("1971-05-14")  
my_date
```

```
"1971-05-14"
```

```
class(my_date)
```

```
"Date"
```

Create Data objects

```
my_date <- as.Date("1971-14-05")
```

```
Error in charToDate(x) :  
character string is not in a standard unambiguous format
```

```
my_date <- as.Date("1971-14-05", format = "%Y-%d-%m")  
my_date
```

```
"1971-05-14"
```


Create POSIXct objects

```
my_time <- as.POSIXct("1971-05-14 11:25:15")  
my_time
```

```
"1971-05-14 11:25:15 CET"
```

Date arithmetic

```
my_date
```

```
"1971-05-14"
```

```
my_date + 1
```

```
"1971-05-15"
```

```
my_date2 <- as.Date("1998-09-29")  
my_date2 - my_date
```

```
Time difference of 10000 days
```

POSIXct arithmetic

```
my_time
```

```
"1971-05-14 11:25:15 CET"
```

```
my_time + 1
```

```
"1971-05-14 11:25:16 CET"
```

```
my_time2 <- as.POSIXct("1974-07-14 21:11:55 CET")  
my_time2 - my_time
```

```
Time difference of 1157.407 days
```

Under the hood

my_date	my_time
"1971-05-14"	"1971-05-14 11:25:15 CET"
unclass(my_date)	unclass(my_time)
498	43064715
	attr(,"tzone")
	" "

Dedicated R Packages

- lubridate
- zoo
- xts

Let's practice!

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